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FROM EMPIRICAL DATA TO INTER-INDIVIDUAL INTERACTIONS: UNVEILING THE RULES OF COLLECTIVE ANIMAL BEHAVIOR

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Animal groups represent magnificent archetypes of self-organized collective behavior. As such, they have attracted enormous interdisciplinary interest in the last years. From a mechanistic point of view, animal aggregations remind physical systems of particles or spins, where the individual constituents interact locally, giving rise to ordering at the global scale. This analogy has fostered important research, where numerical and theoretical approaches from physics have been applied to models of self-organized motion. In this paper, we discuss how the physics methodology may provide precious conceptual and technical instruments in empirical studies of collective animal behavior. We focus on three-dimensional groups, for which empirical data have been extremely scarce until recently, and describe novel experimental protocols that allow reconstructing aggregations of thousands of individuals. We show how an appropriate statistical analysis of these large-scale data allows inferring important information on the interactions between individuals in a group, a key issue in behavioral studies and a basic ingredient of theoretical models. To this aim, we revisit the approach we recently used on starling flocks, and apply it to a much larger data set, never analyzed before. The results confirm our previous findings and indicate that interactions between birds have a topological rather than metric nature, each individual interacting with a fixed number of neighbors irrespective of their distances.

Keywords: Collective behavior; flocking; statistical methods.

AMS Subject Classification: 22E46, 53C35, 57S20

1. Introduction

Collective behavior is a widespread phenomenon in animal groups, which has fascinated scientists for long. Bird flocks, fish schools, insect swarms and mammal herds are just a few common examples where large groups of individuals manage to remarkably coordinate, exhibiting extraordinary collective patterns. In many of these instances, there is no leader or external stimulus to guide the group, rather coordination occurs spontaneously as a consequence of the mutual interactions between individuals — a mechanism known as self-organization.[1]

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How does self-organization emerge in a population of simple interacting individuals? What are the dynamical rules followed by individuals, which lead to global coordination? These questions are central in behavioral ecology, but are also relevant in other areas of science where self-organized collective phenomena have been observed, including social science,[2] economics,[3] control theory,[4] artificial intelligence and cooperative robotics.[5] From this point of view, animal groups represent paradigmatic examples of complex behavior and have attracted a growing multi-disciplinary interest. On the one hand, understanding the mechanistic aspect of group formation can offer a precious background to address cases where individuals are behaviorally more complex and the relevant space highly multi-dimensional (pertaining, e.g. the action strategies of a social agent rather than the coordinates and velocities of a moving bird). On the other hand, collective animal behavior often has important biological functions (e.g. anti-predatory, foraging, mating), the group fulfilling collective tasks that go well beyond the abilities of individuals. In this respect, it represents a powerful inspiration to develop new schemes of distributed coordination, so as to achieve optimal control of artificial systems of interacting agents.

From a closer perspective, complex systems have been the privileged arena for much of frontier research in statistical physics and applied mathematics. Collective behavior, in particular, is a fundamental concept in physics, being at the core of phase transitions in condensed matter. The emergence of order and the progressive onset of long-range correlations have been deeply investigated, providing with sophisticated theoretical approaches, experimental protocols and numerical techniques. This methodology represents a powerful tool of investigation and physicists have become progressively interested in applying the physics framework to problems of interdisciplinary origin. Animal collective behavior represents a rather challenging context in this respect. In the last decade, several groups dedicated intense effort to develop numerical models and theoretical approaches to flocking behavior.[6] The whole area of active matter (see Refs. 7, 8 and Refs. 9, 10 for a more mathematical approach), which focuses on collective movement in systems with active components (from driven granular matter to bacterial suspensions and animal groups), goes back to simple flocking models of self-propelled particles.[11,12]

While the methodology of statistical physics has been largely exploited in theoretical studies, the same is not true for what concerns empirical investigations. Still, as we shall discuss, physics provide a precious background to address several aspects of empirical research on animal groups. As a matter of fact, empirical studies have been limited for long to rather small aggregations (few tens of individuals). This lack of large-scale data has severely restrained the feedback between theories and experiments. As a consequence, models of self-organized motion have not been adequately tested and many of their assumptions, even if reasonable, remained at a speculative level. In this paper we will discuss how and why exporting the physics methodology to empirical research on animal groups can remarkably improve this state of the art, both concerning experimental techniques and data analysis. We will argue that an appropriate statistical analysis on large-scale data allows inferring important properties on

the interactions at play between individuals in a group, providing crucial information for further modeling. To this aim, we will focus on a specific example, and revisit our recent results on the topological nature of the interactions between birds in starling flocks.[13] We will consider a larger data set (never analyzed before) and discuss in great details the methods that can be used to investigate the flocks structure and unveil the topological character of interactions.

2. Exporting the Physics Methodology to Animal Collective Behavior

There are several reasons why physics can provide a useful reference framework for understanding and analyzing collective animal behavior. Even if they have been invoked in several works on flocking, it may be useful to briefly discuss a few of them.

- From a very general perspective, collective animal behavior is qualitatively different, with emerging complex patterns, only when the number of individuals is large. In this respect, animal groups truly represent complex systems, in the sense so well stigmatized in the seminal paper "*More is different*" by Phil Anderson.[14] The behavior of the system as a whole cannot be directly inferred from the knowledge of the individual components, and small aggregations are not informative about collective behavior in large groups. In fact, some of the most stunning and peculiar collective phenomena, like panic waves in bird flocks and fish schools,[15,16] or long-range correlations in orientational behavioral traits,[17] only arise at large enough scales. Collective animal behavior therefore has a deep statistical meaning and considering large groups is a necessary prerequisite to truly understand it. Physicists are used to deal with large systems, and the whole approach of Statistical Physics precisely focuses on characterizing large-scale behavior and deducing it from micro-interactions.
- From a mechanistic point of view, emergence of self-organized collective behavior from local rules is similar to ordering in physical systems of particles or spins with short-range interactions. In animal groups, individuals are thought to obey simple behavioral rules based on imitation: they adjust their velocities to those of neighbors to coordinate their motion, are attracted to them not to lose the group, and keep minimum distances to avoid collisions.[6] Local alignment of orientational degrees of freedom, and attraction–repulsion forces in space are known to occur in several systems in condensed matter. In ferromagnetism, for example, neighboring spins tend to align to each other giving rise to an ordered state at low enough temperature. Interestingly, ferromagnetic systems are considered the paradigm of order–disorder transitions, for which the theory of critical phenomena has been developed in the last five decades. Much is known on how to model them, and characterize large-scale collective properties. On the other hand, also attraction–repulsion forces are well-known in particle systems, and have been widely addressed in liquid theory and hydrodynamics. In animal groups, orientational degrees of freedom (the velocities) are coupled in a nontrivial way with structural ones (the

positions), as individuals continuously diffuse while coordinating. Besides, animals are self-propelled and can move exploiting internal sources of energy, which makes the problem intrinsically of nonequilibrium. Still, the analogies with the physical systems mentioned above are strong. In fact, some of the most successful models of self-organized motion, the so-called self-propelled particle models,[11] have been developed and studied using techniques from statistical physics and exploiting the reference to ferromagnetic and particle systems.[7]

- One of the most peculiar traits of animal collective behavior is the way groups react to perturbations. Bird flocks, for example, sustain frequent predatory attacks retaining a remarkable degree of coherence and cohesion. A similar resilience to external disturbances can be observed in many other species and circumstances (e.g. predatory attacks, physical obstacles, rapid environmental changes, etc.). This collective efficiency requires mechanisms ensuring fast information transfer and structural robustness. Explaining how this might occur is crucial to understand the evolutionary success of grouping and for applications in distributed artificial systems. In physical systems response to perturbations is one of the main subjects of theoretical and experimental investigation. There are powerful concepts and well-established protocols, which can be exported to describe, quantify and interpret responsive behavior in animal groups.

Given these premises, we wish to proceed along the lines of this interdisciplinary tradition, and bring a new contribution in empirical research on collective animal behavior.

3. Empirical Analysis

Collecting data on animal groups can be a formidable task, especially when dealing with 3D aggregations. Insects moving in two dimensions are small and can be kept under laboratory control even in large numbers. The same is true when working at the micro-scale, as with bacteria suspensions. On the contrary, when dealing with animals moving in three dimensions, individuals often have larger sizes and naturally move in a much larger environment. Laboratory control can be problematic, while techniques for field observations may prove extremely complicated. All empirical analysis previous to our work on bird flocks, fish schools and insect swarms were in fact limited to very small groups (few tens of individuals).[18,19] The reason was essentially a technical one. Gathering empirical data corresponds to retrieve the 3D coordinates and velocities for each individual in the group. Optical techniques and mainly stereoscopy — are the way to do that. The main idea is rather simple: multiple and synchronous photographs of the same object are taken, and stereometric formulas are used to retrieve from the set of 2D coordinates on the photographs the 3D coordinates in real space. When dealing with multiple objects, however, one needs first to find the correct correspondence between the projected images of the same object on the different photographs. This matching problem can become extremely severe when

groups are large and compact: photographic images of bird flocks or insect swarms, for example, are typically very dense sets of featureless points. In this case, standard computer vision techniques are inadequate to solve the stereoscopic matching because (i) they require levels of noise unattainable with current experimental technology; (ii) they use probabilistic methods computationally nonexploitable with large numbers. Our approach was rather different. Matching problems are well studied in statistical physics and we largely exploited our background. Several assignment algorithms exist, that can find the optimal matching given a measure in the space of all possible matchings. For the stereoscopic case the crucial point is to find the good measure. To this end, we used a pattern recognition principle, together with the available information on the stereometric transformation. We developed a recursive algorithm providing at each step a better measure, which was eventually integrated with an assignment module, leading to the final match and the resulting 3D reconstruction. Tests on synthetic data indicate an average efficiency of 90% (number of correct matches) on sets of several thousands points. Our aim here is not to describe in details the reconstruction algorithm (the interested reader can refer to Ref. 20), but to underline the crucial role that statistical physics played to solve the stereoscopic matching. Thanks to this new algorithmic methodology, and to an experimental apparatus optimally designed and calibrated (see Ref. 20), we could retrieve the 3D coordinates of starling flocks of up to four thousand individuals, making a leap forward of two orders of magnitude beyond the previous state of the art.[13,21] Here, we will focus on data analysis and revisit some of the techniques that we developed to investigate the structural properties of large aggregations

4. Looking for the Rules

Some of the main questions in collective animal behavior concerns the behavioral rules obeyed by individuals during collective motion. Models assume dynamical rules, which are local, each individual interacting only with a few neighbors. However, very little is known about the real nature of such rules. Who are the interacting neighbors? What are the forces between them? How do they depend on mutual distances? What is the role of attraction, repulsion and mutual alignment? In this respect, empirical analysis can prove crucial. In fact, inferring information on the rules from empirical data is one of our major objectives.

In statistical physics, going back from measurements to the unknown interactions is known as the inverse problem. Some powerful and systematic approaches have been developed to address the inverse problem and have been recently applied to several biological networks. Maximum entropy models are probably the most successful. In this approach, given a set of measurements, one looks for the set of inter-individual interactions that maximizes the statistical occurrence of those measurements. The general theory requires a good ensemble of independent observations to be able to retrieve the underlying interactions. The basic idea is that enough empirical information should be available as an input, to fix all the unknown

interaction pairs (or higher-order links). In fact, the method was successfully applied to neurons in the vertebrate retina, where experiments provided correlation functions for several samples and uncorrelated times.[22] Unfortunately, the same approach is not strictly possible for animal groups. In the case of flocking, for example, different flocks cannot be treated as different samples of the same system. Besides, individuals move through the aggregation, and the interaction network continuously changes in time. Thus, even for a single flock, measurements at different times do not correspond to the same statistical ensemble (i.e. are not macroscopic patterns generated by the same set of bird−bird interactions). Hopefully, however, additional information on the nature of the interactions can be exploited to reduce the complexity of the search space. As a starting point, therefore, we follow an alternative route. We leave aside for a successive stage the possibility of retrieving the detailed interaction between any two pairs of birds, and rather look for some more general property of the interactions, which can be more easily inferred from our empirical data. As a first step, we need to identify a strong tracer of the interactions. As birds in a flock are strongly interacting, we expect their structure in space and dynamical behavior to be different from a system of non-interacting individuals. It is likely that certain quantities will be more affected than others. We will focus on those quantities, and try to extract from their behavior information on the interactions.

4.1. *The data set*

Before proceeding with the statistical analysis, let us briefly discuss the empirical data set we have at our disposal. We performed stereoscopic experiments on starling flocks during aerial display over the roost, for two consecutive years (from mid-October to mid-March in 2005−2006 and 2006−2007). High-resolution digital photographs were shot at 10 frames per second, for a maximum of 40 consecutive images (due to memory constraints of the cameras). The details of the experimental procedure can be found in Ref. 20. Stereoscopic images of selected flocking events were eventually processed with our algorithmic procedure. In this way, we obtained the 3D reconstruction of 19 flocking events. For each event, we have the space coordinates and velocities of individual birds at each instant of time for up to 4 s. Remarkably, the reconstructed flocks are rather diversified in terms of the number of individuals (from a few hundred to almost four thousand), dimensions, velocity and density (see Table 1). This turns out to be crucial to unveil some important features of the inter-individual interactions. A partial analysis of the structural properties of the events of the first season (10 flocks) appeared in Ref. 13. Here, we will consider the complete data set, which includes the largest flocks reconstructed so far (see Table 1).

4.2. *The problem of the border*

When investigating real flocks, some features occur that are usually absent in theoretical analysis and numerical investigations. Flocks, for example, have a well-defined border. On the one hand, the border represents a truly relevant biological

Table 1. Main global features of the analyzed flocking events. Each value reported in this table is an average over all the instants of time (i.e. 3D reconstructions) belonging to that event. Events are numbered according to the session of data-taking (the first number indicates the day of the stereoscopic experiment, the second the photographic series within that day). The last nine events were reconstructed in 2006−2007 and were never analyzed before.

Event	Number of birds N	Density ρ (m^{-3})	Nearest-neighbor distance r_1 (m)	Velocity V (ms^{-1})
16-05	2631	0.081	1.25	15.2
17-06	534	0.08	1.17	9.1
21-06	617	0.24	0.92	11.2
25-08	1360	0.09	1.14	11.9
25-10	834	0.34	0.82	12.0
25-11	1168	0.38	0.75	8.8
28-10	1246	0.54	0.67	11.1
29-03	448	0.13	0.99	10.1
31-01	1856	0.04	1.44	6.9
32-06	781	0.8	0.68	9.6
48-17	837	0.12	0.96	9.5
49-05	755	0.84	0.6	13.8
54-08	4150	0.08	1.21	13.7
57-03	3070	0.09	1.15	12.3
58-06	428	0.45	0.73	10.2
63-05	855	0.15	1.02	9.9
69-10	1108	0.12	1.06	11.9
69-13	1861	0.15	1.03	11.7
69-19	763	0.37	0.7	13.3

property, which presumably has a strong influence on several collective patterns. From this point of view, some of the most sophisticated theoretical analyses on self-propelled particles remain poorly informative on animal groups, as they focus on fluids of particles rather than finite aggregations, do not take into account the presence of a boundary, or assume periodic boundary conditions. On the other hand, the presence of the border requires particular care during data analysis: birds on the border have different statistical properties simply because part of the space around them is empty. Appropriate methods must be used to treat this statistical bias and disentangle the effect of this positional anomaly from the effect of the interaction between individuals.[23] Therefore, the border of the flock must be accurately identified. As flocks are not necessarily convex, the so-called convex hull is not a suitable tool. Rather, to define the border we used the α-shape algorithm[24]: basically one excavates the set of 3D points (i.e. birds) with spheres of radius α. For $\alpha = \infty$ the border coincides with the convex hull, whereas for finite α concavities of size α are detected. More specifically, the scale α defines a sub-complex of the Delaunay triangulation (the α-complex) including all the simplices, which have an empty circumsphere with squared radius equal to or smaller than α. The external surface of this triangulation, the α-shape, identifies the border (see Fig. 1). In our analysis we choose the scale α as to locally maximize the density of the set, as shown in Fig. 2.

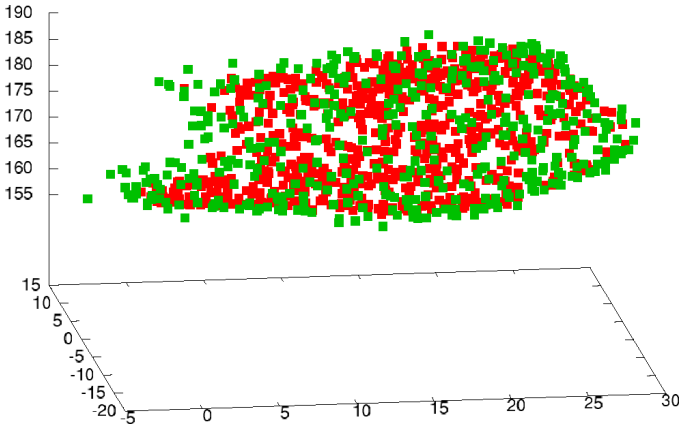


Fig. 1. (color online) Three-dimensional reconstruction of flock 69-10, 1108 birds: red points correspond to internal birds, while green points — obtained using the alpha-shape algorithm — identify birds on the border.

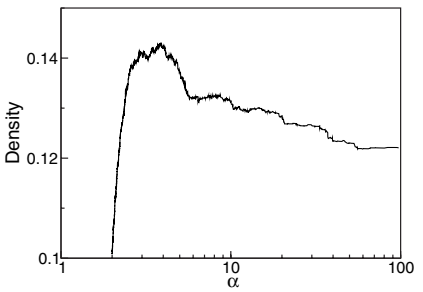


Fig. 2. Density of internal points as a function of the parameter α, for the flock of Fig. 1. The density is computed dividing the number of internal birds (i.e. birds that do not belong to the border) by the volume enclosed in the α-shape. When α is infinite, the α-shape coincides with the convex hull. As α decreases, the algorithm excavates the set of points with a higher resolution. If concavities are present in the shape of the flock, when α is of the order of the concavities scale, the α-shape volume suddenly decreases, giving rise to a jump in the density. At small values of α, the algorithm penetrates the flock, progressively eroding points and leading to a decrease in density. The optimal choice is therefore to consider the value of α where the density is maximal.

This corresponds approximately to considering the smallest concavities without penetrating inside the flock. Boundary effects in all statistical indicators are then cured using the Hanish method.[25]

4.3. *The anisotropy*

Statistical physics teaches us that the effect of interactions is mostly seen in correlation functions and two-point observables. In the case of flocks, we can consider either structural properties (describing how individuals are distributed in space) or orientational ones, (pertaining to the degree of mutual alignment). Here, we focus on

structural features and leave to the final section a few comments on orientational degrees of freedom.

One of the simplest quantities that can be computed with a set of 3D coordinates is the nearest-neighbor distribution. Given a reference bird i, we consider the nearest-neighbor ($1n$) and compute its distance $d_i = |\mathbf{r}_{1n} - \mathbf{r}_i|$ and normalized displacement $\mathbf{u}_i = (\mathbf{r}_{1n} - \mathbf{r}_i)/d_i$ from i. Looking at all the birds in the group, we can measure the distribution of absolute distances and angular displacements of nearest neighbors. This last quantity appears to be particularly interesting. Let us consider as a reference direction the (normalized) average velocity of the flock $\mathbf{V} = \sum_i \mathbf{v}_i / |\sum_i \mathbf{v}_i|$, and measure the distribution of $\cos(\theta) = \mathbf{u}_i \cdot \mathbf{V}$. The result is shown in Fig. 3(a) for a

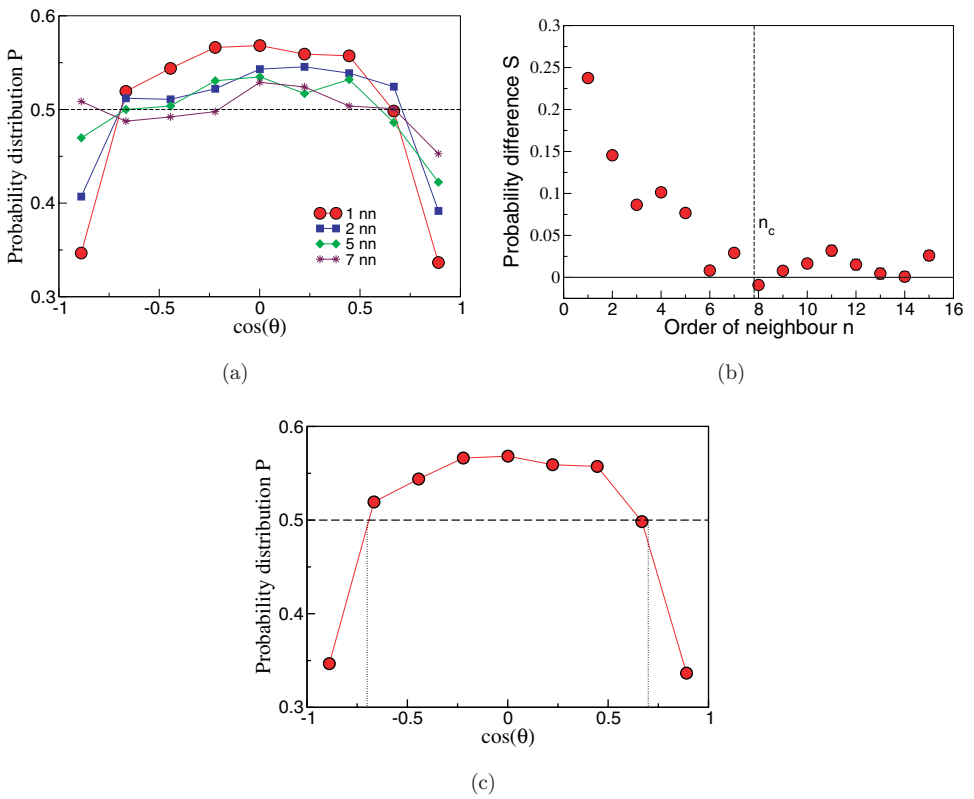


Fig. 3. (color online) (a) Probability distribution P of the cosine of the angle between the nearest-neighbor vector and the direction of motion, for flocking event 69-10. The curve is an average over all the instants of time belonging to the event. The black horizontal line corresponds to the isotropic case, when neighbors are uniformly distributed around the focal individual. The dotted vertical lines identify the reference angles used in the computation of the probability difference S (see text and (c)). (b) Probability distributions for the first (red), second (blue), fifth (green) and seventh (brown) nearest neighbors. (c) Relative probability difference S of finding the nth neighbor below a reference angle with the direction of motion, as a function of the order of the neighbor n. To compute S, we identify the range where the probability P is lower than the isotropic case (e.g. $|\cos(\theta)| < 0.7$ in this example, see vertical lines in (a)) and calculate in this range the relative integrated probability difference with the isotropic case. The value where the curve drops below 0, identifies the interaction range.

single flocking event. We can see that there is a significant drop in probability for values of $\cos(\theta)$ close to $+1$ and -1. In other terms, it is rather unlikely to find the nearest neighbor along the direction of motion.

This anisotropy in the angular distribution of neighbors is necessarily a consequence of the interaction at play between birds: in a system of non-interacting individuals, neighbors would be isotropically distributed in space and $P(\cos(\theta))$ would be constant and equal to $1/2$. If we repeat the same analysis for the second nearest neighbor, a similar distribution is obtained, but with a less pronounced anisotropy. Increasing the order of the considered neighbor, $P(\cos(\theta))$ progressively becomes flatter: in the event of Fig. 3(a), for the seventh nearest neighbor P is very close to an isotropic distribution of points (Fig. 3(b)). As anisotropy is a consequence of the underlying interactions, this finding indicates that (i) interaction is strong between a bird and its closest neighbors so as to significantly modify their mutual positioning; (ii) interaction decays as the order of the neighbor increases. Interestingly then, if we are able to quantify this decay, we can estimate the range of the interaction and investigate how it depends on some *a priori* relevant parameters of the flock (density, velocity, dimensions etc.). This would provide novel and important information on the dynamical rules followed by individuals.

4.4. *Estimating the range of the interaction*

To estimate the range of the interaction, we need a sharp statistical indicator of the degree of anisotropy. As a first possibility, we can consider $P(\cos(\theta))$ and quantify how much it differs from the isotropic value $1/2$: the larger this difference, the larger the anisotropy. This can be done, for example, by measuring the relative probability difference of finding a neighbor below some reference angle, as compared to the isotropic case (Fig. 3(a)). If we repeat the procedure for different orders n of the considered neighbor, we obtain the curve $S(n)$ displayed in Fig. 3(c): as expected $S(n)$ decays as n increases, reaching the non-interacting value (0 in this case) for a certain value n_c, that we can identify as the interaction range.

$S(n)$ provides an intuitive quantification of the anisotropy. It is not, however, an optimal one: it slightly depends on the reference angles used to evaluate the integral, and especially in small aggregations it suffers of large relative fluctuations. It would be desirable to find more robust statistical indicators. To this aim, we introduce a more sophisticated quantity, the Anisotropy Matrix[13]:

$$M^{(n)}_{\alpha,\beta} = \frac{1}{N}\sum_{i}^{N} u_i^{\alpha} u_i^{\beta} \tag{4.1}$$

where, as before, u_i represents the normalized distance vector of the nth nearest neighbor of bird i. The matrix M is the sum over many projectors: the operator indeed projects over the direction of the nth nearest neighbor of i. As a consequence, the eigenvectors of M have a particular meaning. If we call $\lambda_1 < \lambda_2 < \lambda_3$ the three

eigenvalues of M, and $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ the corresponding eigenvectors, then $\mathbf{w}_1$ and $\mathbf{w}_3$ represent, respectively, the directions where the nth nearest neighbor of a reference bird is statistically less and more likely to be found. This property can be exploited to quantify the anisotropy in the neighbors distribution and monitor its decay.

From the analysis of $P(\cos(\theta))$ we know that nearest neighbors are less abundant along the direction of motion $\mathbf{V}$ (Fig. 3). Therefore, we expect the lowest eigenvector $\mathbf{w}_1$ to be quite similar to $\mathbf{V}$ and their scalar product to be rather close to 1. When the order of the considered neighbor increases, the anisotropy is quantitatively less pronounced so that this scalar product must become smaller. The squared scalar product $\gamma_1(n) = (\mathbf{w}_1 \cdot \mathbf{V})^2$ seems therefore a good candidate to measure the degree of anisotropy. To this aim, however, it is important to establish first what is the value this function would have when individuals are not interacting and the distribution of neighbors is isotropic. In this case, the three eigenvectors $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ are only determined by finite size fluctuations and are statistically uncorrelated with $\mathbf{V}$, as there is not any *a priori* privileged direction in space. The average value of $\gamma_1(n)$ can be easily computed by assuming $\mathbf{w}_1$ to be uniformly distributed on the unitary sphere. The same holds when considering the gamma functions $\gamma_2(n)$ and $\gamma_3(n)$ associated with the other two eigenvectors, and one gets for the isotropic case: $\gamma_1^{\text{iso}}(n) = \gamma_2^{\text{iso}}(n) = \gamma_3^{\text{iso}}(n) = 1/3$.

The behavior of $\gamma_1(n)$ as a function of n is shown in Fig. 4 (red curve) for three flocking events. This curve is in fact an average: we diagonalized first the anisotropy matrix and computed $(\mathbf{w}_1 \cdot \mathbf{V})^2$ for each single-time 3D reconstruction, and we then averaged over all the instants of time partaking the event. The function $\gamma_1(n)$ exhibits, as expected — a neat decaying behavior from a large value at $n = 1$ to the isotropic non-interacting value 1/3. The order of the neighbor where $\gamma_1(n)$ becomes comparable to $\gamma_1^{\text{iso}}(n) = 1/3$ can be identified as the interaction range n_c. To further support this procedure, we can consider the gamma functions corresponding to the other two eigenvectors of the anisotropy matrix, $\gamma_2(n)$ and $\gamma_3(n)$ (Fig. 4, curves green and blue). We recall that the maximal eigenvector of M identifies the direction of maximal crowding. Therefore, the behavior of $\gamma_3(n)$ indicates that nearest neighbors are preferably located on the plane orthogonal to the direction of motion (as also suggested by Fig. 3). This is another effect of the anisotropy in the distribution of neighbors, which progressively fades away as the order of the neighbor increases. All the three gamma functions then cross at $n \sim n_c$ and fluctuate around 1/3 for farther neighbors, consistently with the fact that the distribution of neighbors is isotropic beyond n_c.

The anisotropy matrix and the gamma functions provide a systematic and robust protocol to estimate the interaction range. Focusing on an aggregated quantity (M), we exploited the advantage of dealing with large samples, reducing the effect of bird-to-bird fluctuations. Besides, averaging over measurements at different instants of time we further limited statistical fluctuations due to finite size effects. In this way we could compute a scalar quantity, $\gamma_1(n)$, which on one hand encodes the information on the anisotropy in an efficient and compact way, on the other hand exhibits

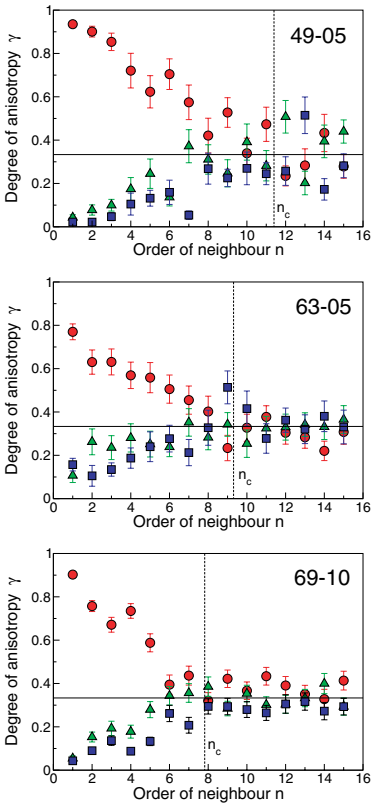


Fig. 4. (color online) Anisotropy gamma factors γ_1 (red circles), γ_2 (green triangles) and γ_3 (blue squares) for three flocking events (49-05; 63-05; 69-10). Each curve is an average over all instants of time belonging to the flocking event, error bars are standard errors. The interaction range n_c is identified as the point where the function γ_1 reaches the isotropic value $1/3$ (black horizontal line), and is computed using a linear fit of the curve close to the intersection. The three gamma functions meet together approximately at n_c, as expected, and fluctuate around the isotropic value.

a smooth decaying behavior. We stress that to achieve such a result, it was crucial dealing with large groups and taking care of boundary effects. Birds on the border or close to it necessarily have their neighbors on one side. This neighbor location is not determined however by the interaction between individuals, rather it is a trivial effect of the shape of the flock (e.g. birds in front of the group have all neighbors behind, while birds on the left side have all neighbors on their right etc.). It is easy to check that even for a random aggregation of points, where neighbors are distributed isotropically, computing $P(\cos(\theta))$ and the gamma functions in a naive way does not give the expected isotropic value.[23] As our aim is to use the anisotropy as a tracer of the interaction, we need to eliminate this bias. Luckily, statistical methods exist to this purpose. We followed the Hanish method[25]: when computing observables relative to the nth neighbors we considered only birds for which the nth neighbor was at a distance smaller than the distance to the border of the flock. In this way one obtains

an unbiased statistics, which can however be much smaller than the original one (especially for large n, when many points must be discarded). For this reason large aggregations are mandatory to obtain significative results.

4.5. *Topological versus metric interactions*

Once established a good statistical indicator and a procedure to measure the degree of anisotropy, we applied it to all the flocking events in our data sets, and computed for each of them the corresponding interaction range n_c. Inspection of the results shows that n_c slightly varies from flock to flock, as can also be seen in Fig. 4. This must come as no surprise. n_c is an estimate of the typical number of interacting neighbors and has been computed using averaged quantities on finite aggregations of points. Any measurement on finite groups is expected to deviate from the true value, with group-to-group fluctuations that are smaller, the larger their sizes. The point is not whether n_c varies from one flock to the other, but whether this change is a natural statistical fluctuation or, rather, is determined by some specific parameter of the flock. The relation with the flocks density, in particular, plays a crucial role. Let us discuss why.

Most models of self-organized motion assume that during collective behavior individuals interact with neighbors within a well-defined range in space. Interactions, in other terms, depend on mutual metric distances. This is what happens in all physical systems and may seem a reasonable assumption even for animal groups, as several perceptual abilities of individuals decay with distance. If this hypothesis is correct, there is an interaction range r_c, which is fixed in meters. In this case, the number of neighbors a bird is interacting with crucially depends on the density of the group: in dense aggregations individuals are close to one another and many neighbors will fall within interaction range r_c; on the contrary, in sparse aggregations only a few neighbors will satisfy this condition. Thus, the number of interacting neighbors n_c should increase with increasing density. This is not, however, the only possibility. It is well known that in many social networks it is not the physical distance in space between nodes that matters, and other topologies are the relevant ones. In the case of flocks, for example, one may argue that the relevant quantity is not how far apart two birds are (metric distance), but how many intermediate individuals separate them (topological distance). If this topological scenario holds, interaction decays with order of neighborhood and there is a well-defined topological range n_c: each individual interacts with the first n_c neighbors, irrespective of their distances. In this case the number of interacting neighbors n_c would not depend on density, but their distance would.

Our empirical results can help to distinguish in a sharp way between these two scenarios. The anisotropy matrix allowed us to extract an interaction range in terms of the number of neighbors, n_c, that we may now call topological range. For a given flock, however, the same procedure also provides with a metric interaction range in space, r_c. For homogeneous aggregations — as flocks are, to a good approximation — there is indeed a very clear relationship between the order of the neighbor n and its average

distance r_n:

$$n \propto \frac{4}{3}\pi r_n^3 \rho \ , \tag{4.2}$$

or, more conveniently

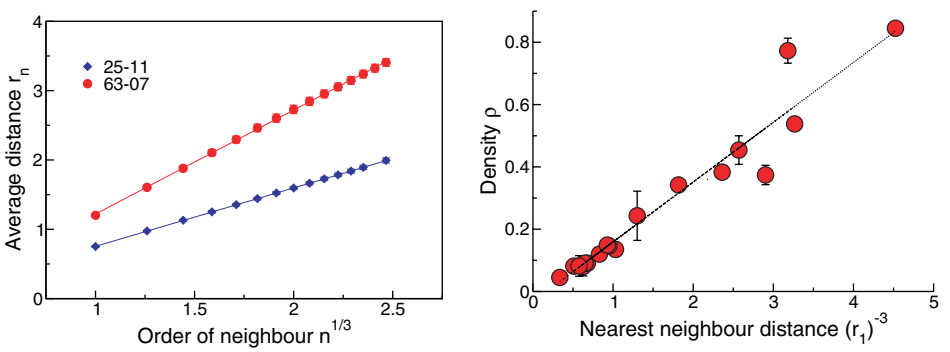
$$n^{1/3} \propto \rho^{1/3} r_n \sim \frac{r_n}{r_1} \ , \tag{4.3}$$

ρ being the density, and $r_1 \sim \rho^{-1/3}$ the average nearest-neighbor distance. Empirical data follow nicely this behavior, as can be seen in Fig. 5. As a consequence, the gamma functions defined in the previous section can be easily expressed in terms of distances, the decay of the anisotropy monitored in physical space, and a metric interaction range r_c identified.

For a single flock, the topological and the metric range are equivalent descriptions of the decay of the inter-individual interaction and are related by the homogeneity relationship

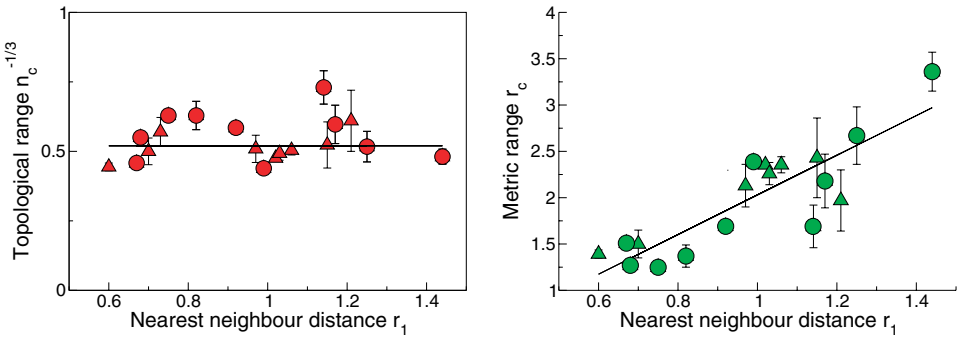
$$n_c^{1/3} \propto \frac{r_c}{r_1} \ . \tag{4.4}$$

When flocks of different densities are considered, however, these two quantities must behave in a different way and cannot be both density-independent — as Eq. (4.4) clearly shows. If the nature of the interaction between birds is metric, then the metric interaction range r_c must remain constant with varying the density, while $n_c^{-1/3}$ should grow linearly with the nearest-neighbor distance r_1. On the contrary, if the interaction is topological, n_c should remain constant, and r_c must grow linearly with



(a) (b)

Fig. 5. (a) Average distance of the nth nearest neighbor as a function of the order of the neighbor $n^{1/3}$, for two flocking events. The straight lines are linear fits of the data (reduced chi-squared 10^{-5}). The Hanish method is used to compute average distances, in order to eliminate the border bias. (b) Density of the aggregation as a function of the average nearest-neighbor distance r_1^{-3}. Each point corresponds to a distinct flocking event and is an average over all instants of time belonging to that event (error bars are standard errors). The straight line is a linear fit of the data (reduced chi-squared $= 4 \cdot 10^{-4}$). The linear correlation between density and r_1^{-3} shows that the average nearest-neighbor distance is a reliable measure of the sparseness of the flock.



(a) (b)

Fig. 6. (a) Topological range $n_c^{-1/3}$ as a function of the average nearest-neighbor distance r_1. There is no significative correlation between these two quantities, rather the topological range fluctuates around the average value $\langle n_c \rangle^{-1/3}$ (black line), where $\langle n_c \rangle = 7.04 \pm 0.6$. (b) Metric range r_c as a function of the average nearest-neighbor distance r_1. In this case an evident linear correlation exists (reduced chi-squared $8 \cdot 10^{-2}$, straight black line). These results indicate that the interaction between birds has a topological nature, each individual interacting with a fixed number of neighbors, irrespective of their distances.

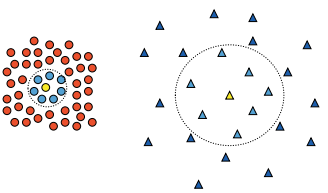


Fig. 7. (color online) Pictorial representation of a topological interaction: the number of interacting neighbors (in light blue) is the same in dense (left) and sparse (right) aggregations; on the contrary the metric range changes, being smaller the denser the flock (dotted circle).

r_1. Our data set comprises flocking events with a wide range of densities and therefore allows one to investigate this issue thoroughly. Figure 6 shows the behavior of $n_c^{-1/3}$ and r_c as a function of the nearest-neighbor distance r_1. The result indicates very clearly that the interaction has a topological nature, supporting the conclusion of Ref. 13, where half of the flocking events were considered in the analysis. Thus, in starling flocks each bird interacts with a fixed number of neighbors irrespective of their distances (see Fig. 7): averaging over all flocking events we get $\langle n_c \rangle = 7.04 \pm 0.6$, improving the previous empirical estimate of this number given in Ref. 13.

5. Conclusions

Our empirical analysis shows that the interaction at play between starlings during flocking has a topological nature, each bird coordinating with a fixed number of interacting neighbors during motion, approximately seven, irrespective of their distances. This finding overturns one of the fundamental hypotheses of most models of self-organized motion, which assumed interactions depending on metric distances.

From a broader perspective, it fosters a number of important questions, which may help to improve our understanding of collective animal behavior and influence future research in this field.

Collective behavior in animals often has functional motivations, being in a group proving advantageous under many circumstances. Understanding how the group acquires its collective skills and which mechanisms grant functional efficiency is a central issue both in evolutionary ecology and in applications to artificial systems. In the case of starlings, as in many other bird species, flocking mainly serves as a strong anti-predatory strategy. In this perspective one must expect that behavioral rules have been evolutionarily selected to enhance those group features that are crucial to this function: a strong cohesion of the group, robust to external perturbations (i.e. resilient to predator attacks); and an equally strong coherence, to change rapidly group direction when necessary. We remain truly impressed when observing a flock dodging a falcon attack: the group undergoes sudden density variations, expanding, contracting, and changing shape and direction. Still, it always remains cohesive, leaving no stragglers behind. Interestingly, metric interactions are not well suited to sustain such behavior. If each bird interacts with neighbors within a fixed metric range, when the flock expands the number of such neighbors drastically decreases, and several individuals may lose connection with the group remaining isolated (and, therefore, easy preys). Topological interactions, on the contrary, do not suffer any dampening with density variations as their ranges do not depend on metric distance. Indeed numerical simulations show that topological interactions make the group much more flexible and resilient to external perturbations, either disruptive as predators, either passive as mere obstacles.[13] Therefore, we can argue that birds in a flock interact topologically because these interactions guarantee a much more robust cohesion to the group, enhancing their chance of survival.

The behavioral rules followed by individuals fully determine the collective features of the group. On the other hand, such rules ultimately depend on complicated biological processes at individual level. The sensorimotor apparatus and the cognitive information processing capabilities of the individuals necessarily constrain the way animals interact with one another, determining the behavioral rules they follow. Understanding how this occurs may not be trivial. In flocking, vision is known to be the main channel through which individuals interact. Still, it may not be vision in itself what matters. In particular, birds perform an elaboration of the visual signal, which goes beyond mere perception, and display prenumeric subitizing abilities (i.e. object tracking). This may play a crucial role in determining behavioral rules. For example, it has been shown in some empirical works that seven is the maximum number of objects that birds can discriminate.[26] We find a very similar value for the number of interacting neighbors. It is possible that the two findings are in fact related, and that each individual considers its first seven neighbors to coordinate with, because this is the maximum number of individuals it can track. Obviously, this hypothesis needs to be investigated further, but the topological nature of the interaction may in fact be related to a nontrivial cognitive ability.

As topological interactions enhance robust cohesion in flocking, we can see that sensory-cognitive constraints on the one hand shape the individual dynamics; on the other hand, they ensure efficient group behavior. In a way, individual cognition scales up to collective information processing skills. It is not evident that topological interactions do play the same role in all other instances of collective behavior. We expect them to occur whenever structural robustness is functional to the group and compatible with individual abilities. Comparative empirical research on different species can help to investigate this issue. Hopefully, we will understand what rules are better suited to achieve certain collective tasks, given specific constraints on group members.

At methodological level, our approach has focused on correlation functions. The anisotropy gamma functions are in fact nothing else but some kind of nonstandard structural two-point functions. What we have done can be rephrased by saying that we measured the correlation length of this anisotropy function and used it as an estimate of the interaction range. This procedure makes perfect sense. Statistical physics teaches us that when the correlation length is finite, it is indeed proportional to the interaction range. Being more cautious, we can say that n_c provides an upper bound for the interaction range, as usually correlations extend beyond the range of direct interactions. The most important point is, however, that this estimated range does not depend on density, unequivocally showing the topological character of the interaction. We note also that n_c does not show any significant correlation with other parameters of the flock, like size, number of birds, velocity and degree of alignment, supporting its identification as a *bona fide* interaction range.

The analysis presented in this paper concerns structural degrees of freedom. We investigated properties relative to the distribution in space of individuals, which are mainly determined by the attraction−repulsion forces between birds. In this respect, the interaction range that we found is, strictly speaking, relative to this structural component of the interaction. Still, an important contribution to the interaction is given by mutual velocity alignment and concerns orientational degrees of freedom. Just as attraction forces are crucial to ensure cohesion in the group, alignment strongly determines the degree of global coordination. Here, we unveiled the topological nature of structural forces, arguing that it ensures robust cohesion, enhancing response to predation. In the same way, we can ask whether orientational degrees of freedom also interact topologically and what kind of alignment interaction can grant efficient global coordination. To answer this question, one can follow the same strategy that we used in this paper: identify a correlation function, which is mostly affected by the interaction and monitor its decay. We recently performed such an analysis in Ref. 17, where we studied velocity correlation functions for the same events discussed in this paper. What we found was startling: velocity correlations in flocks are scale-free and span the entire group with minimal decay. On the one hand, this result tells us that flocks are coherent in the strongest possible way, such that the collective response to perturbations, in terms of orientational changes, is maximal. On the other hand, scale-free behavior implies that correlations do not have a

well-defined length-scale related to the original interaction range: direct inspection of these functions cannot therefore be used to trace back this range, as we did with the anisotropy. More sophisticated approaches, like maximum entropy models integrated with empirically based hypothesis, may provide this information. Work in this direction is underway.

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